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Bichat

Bichat was a French anatomist and physiologist, best remembered as the '**Father of modern histology**' and **pathology**. He coined the term '**tissue**'. He revealed that diseases attack tissues rather than whole organs.

Unit

2

Structural Organisation in Animals

TISSUES-THE 'JACKS OF ALL TRADES AND MASTERS OF NONE'

In the evolution of animals, the major event of significance is the 'development of tissues'. The '**tissue grade of organisation**' evolved for the first time in the '**cnidarians**'. When the zygote undergoes cleavage, it results in the production of many cells. During the embryogenesis itself, cells differentiate themselves into different kinds of tissues to perform different functions (**division of labour**). When a cell is marked for a specific function, both the cell and its descendants are destined to do that specific function only (except in the **sponges**). In the parazoans, mature cells can '**dedifferentiate**' or 'redifferentiate' and perform certain functions that are not their usual functions. For example, a choanocyte of a sponge can stop its normal function of collection of food and digestion and can produce gametes.

Production of tissues has an *evolutionary advantage* compared to organisms in which there are no '**true tissues**' (e.g., **sponges**). As somebody rightly observed '**Tissues are jacks of all trades and masters of none**' (they do take part in every biological activity doing their assigned function, contributing their part for the completion of the overall **integrated biological activity** of the organism). Tissues are essentially of **four** types – epithelial, connective, muscular and nervous.

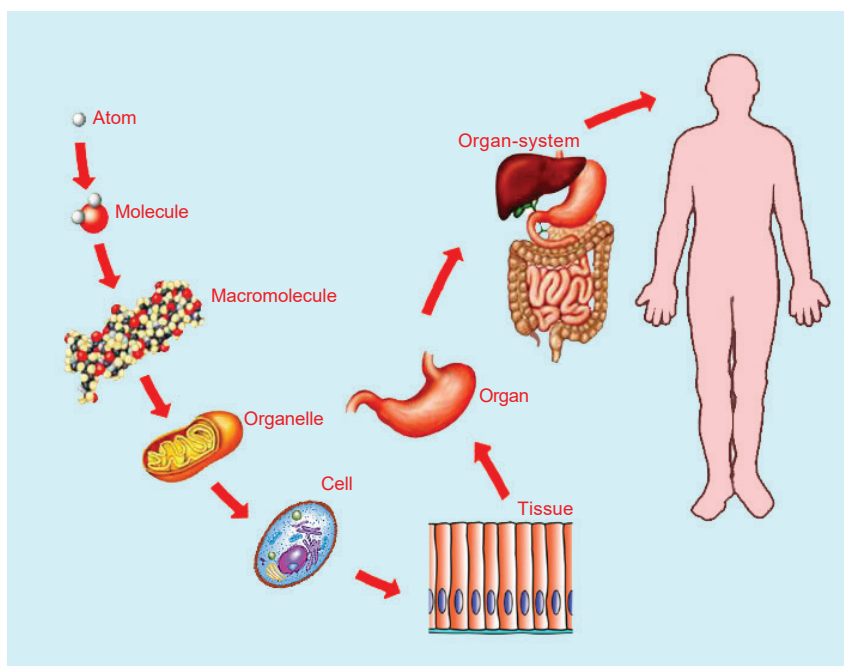
Epithelial tissues cover the outer surfaces/free surfaces of the body organs and act as 'physical barriers' protecting organisms from pathogens. They are also involved in the **repair of injuries**. Epithelia in 'sense organs' receive **sensory stimuli**. They form '**glands**', which are **secretory** in function, in certain regions of the body.

Tissues show several adaptations to meet the requirements of the body. **Connective tissues** help in binding and supporting other tissues. A connective tissue has diffuse cells interspersed by 'extracellular matrix' made up of different types of fibres (collagen, elastic and reticular fibres). Various types of fibres are formed by cells called '**fibroblasts**'. The matrix of connective tissue also contains '**macrophages**' which 'phagocytize' pathogens.

Muscular tissues are concerned with various types of movements in the body. They contain proteins such as '**actin**' and '**myosin**', which together cause contraction in a muscle. Muscles are of three types - skeletal, cardiac and smooth. *Skeletal* and *cardiac* muscles are of the '**striated**' type and smooth *muscles* (also called **visceral muscles**) are '**unstriated muscles**'.

Nervous tissue senses 'stimuli' and conducts (**transmits**) them in the form of **nerve impulses** (*action potentials*) from one part of the body to another. It consists of nerve cells called '**neurons**'. Nervous tissue also contains various types of '**glial cells**', which help in nourishment, insulation of axons, protection etc. Nervous tissue is chiefly involved in bringing about **coordination** between the activities of various body parts (**neural integration**).

You probably, understand now why somebody called tissues the '**JACKS OF ALL TRADES BUT MASTERS OF NONE**'.



Levels of organisation

2.1 Levels of Organisation

2.2 Symmetry

2.3 Coelom

2.4 Animal tissues

2.1 LEVELS OF ORGANISATION

Living world includes a large number of acellular and multicellular organisms with vast differences in their structure and form. In spite of the differences in the structure and form of different animals, there are fundamental features common to various individuals in relation to the arrangement of cell layers, symmetry and nature of coelom, patterns of digestive, circulatory, reproductive systems, etc.

There are five levels of organisation from the protozoans to the higher metazoans. They are as follows:

Protoplasmic level

Protozoans are the simplest protists. All the basic functions of life such as locomotion, digestion, respiration, excretion, osmoregulation and reproduction are performed by a single cell (The single cell acts as a complete organism) in protozoans. Within the cell, protoplasm is differentiated into cell organelles which carry out different functions. It is called **protoplasmic level of organisation**.

Note: Mention of the protoplasmic level is only to provide better understanding as only metazoans are dealt with in this chapter.

2.1.1 Levels of organisation in the metazoans

Though all the members of the kingdom Metazoa are multicellular, all of them do not exhibit the same pattern of organisation of cells. Cells of the metazoans are not capable of independent existence, and exhibit 'division of labour' at various levels of organisation. Among the metazoans of different kinds, cells may be functionally isolated or grouped together to form tissues, organs or organ-systems and as such structural complexity increases from the lower metazoans to the higher metazoans.

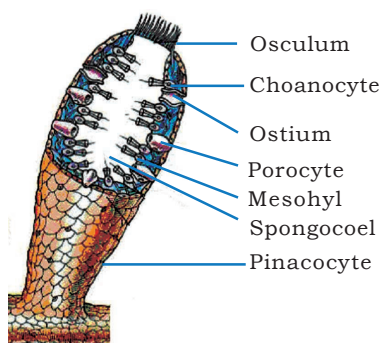


Fig. 2.1 Sponge (body partly cut open) showing cellular level of organisation.

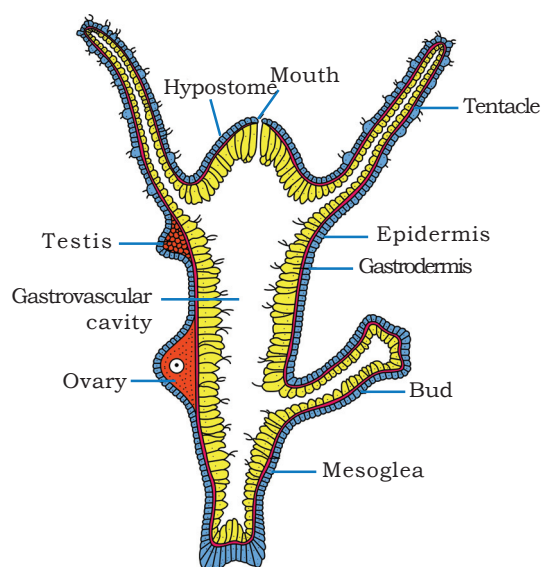


Fig. 2.2 L.S. of Hydra (Showing tissue level of organisation)

The levels of organisation in metazoans are as follows:

I. Cellular level of organisation

It is the lowest level of organisation among the metazoans and is exhibited by the sponges (parazoans). Different types of cells are functionally isolated due to the absence of sensory and nerve cells. The cells are arranged as '**loose cell aggregates**' and do not form tissues. There is division of labour among the cells. The outer layer of body wall is formed by pinacocytes and porocytes (cells that conduct water from the outside into the spongocoel), whereas the inner layer lining the spongocoel is formed by choanocytes. In between the two layers, a non-cellular mesohyl is present.

II. Tissue level of organisation

This is the lowest level of organization among the eumetazoans, exhibited by **diploblastic** animals like the cnidarians. In these animals, the cells which perform the same function are arranged into **tissues**. The cells of a tissue together perform their common function as a highly coordinated unit and this coordination is due to the presence of **nerve cells** and **sensory cells**. **The formation of tissues is the 'first key transition' in the evolution of the animal body plan.** Development of bilateral symmetry is the 'second key transition' and the origin of perivisceral cavity (body cavity) is the 'third key transition'.

III. Organ level of organisation

An aggregation of different kinds of tissues which is specialized for a particular function is called an organ. Organ level of organisation is a further advancement over the tissue level in the evolution of levels of organisation. Organ level of organisation appeared for the first time in the members of the Phylum Platyhelminthes. (Ref: NCERT XI Class biology text book)

IV. Organ-system level of organisation

It is the highest level of organisation among the animals and is exhibited by the triploblastic animals such as the flat worms, nematodes, annelids, arthropods,

molluscs, echinoderms and chordates. In the triploblastic animals, the evolution of '**mesoderm**' resulted in structural complexity. In these animals, the tissues are assembled to form organs and complex organ-systems. Highly specialized sensory and nerve cells bring about a higher level of coordination and integration among the various organ systems to lead an efficient way of life. Organ systems in different groups of animals exhibit various patterns of complexities. For example, the alimentary canal in the members of Platyhelminthes has only a single opening to the exterior, and it serves as both mouth and anus, hence it is called **incomplete gut**. A complete alimentary canal has two openings, the mouth and the anus. Such a gut is seen in the animals ranging from the nematodes to chordates. Similarly, circulatory system may be of two types.

- a) **Open type:** In this type, blood pumped out of the heart flows through open spaces bathing the cells and tissues directly. e.g., members of Arthropoda, Mollusca, Echinodermata and Urochordata.
- b) **Closed type:** In this type, the blood is circulated through a series of vessels of varying diameter (arteries, veins and capillaries.) e.g., annelids, cephalopods, cephalochordates and vertebrates.

During the evolutionary history of animals, there has been a tendency towards increase in the body size, explain Why?

2.1.2 Diploblastic organisation

Animals, in which the cells are arranged in two embryonic layers, the external **ectoderm** and the internal **endoderm**, are called **diploblastic animals**, e.g., cnidarians and ctenophores. Ectoderm gives rise to the **epidermis**, the outer layer of the body wall. Endoderm gives rise to the **gastrodermis**, the tissue that lines the gut cavity. The cells in each tissue layer are functionally interdependent. An undifferentiated layer, present between the ectoderm and endoderm, is the '**mesoglea**'. In some diploblastic animals, cells occur in the mesoglea, but they are always derived either from the ectoderm or from the endoderm.

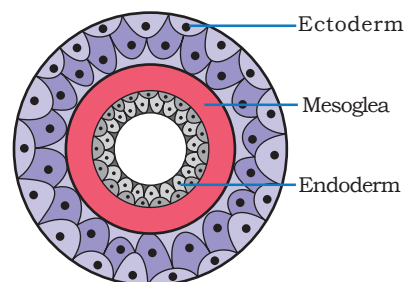


Fig. 2.3 (a) Diploblastic level of organisation

2.1.3 Triploblastic organisation

The animals, in which the developing embryo has a third germinal layer, the **mesoderm**, in between the ectoderm and the endoderm, are called **triploblastic animals**, e.g., flat worms to chordates. Most **triploblastic** animals have an organ-system level of organization. Tissues are organized into organs which form excretory, nervous, digestive, reproductive, circulatory and other systems. Triploblastic animals are usually **bilaterally symmetrical**. (Recall – the taxonomic grade Bilateria is also called Triploblastica).

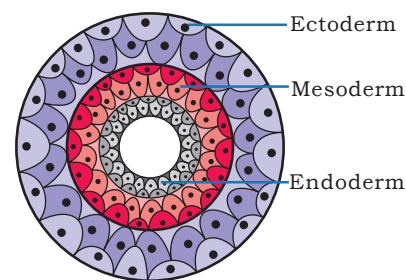
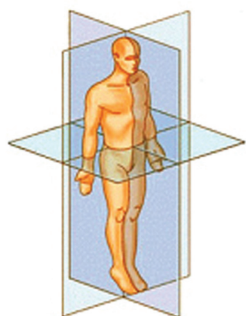


Fig. 2.3 (b) Triploblastic level of organisation



Human body showing symmetry



Sponge showing asymmetry

2.2 SYMMETRY

The concept of symmetry is fundamental in understanding the organisation of an animal. Symmetry in animals is balanced distribution of paired body parts. The body plan of a vast majority of metazoans exhibits some kind of symmetry. However, most of the sponges and snails show asymmetry (lack of symmetry). The symmetry of an animal and its mode of life are *correlated*.

2.2.1 Asymmetry

The animals, which cannot be cut into two equal parts (antimeres) in any plane passing through the centre of the body are called asymmetrical, e.g. most sponges and adult gastropods. In the asymmetrical animals, the body lacks a definite form. Asymmetry cannot be said to be an adaptation or advantage to an organism. Primarily asymmetrical organisms do not develop complex sensory and locomotor functions.

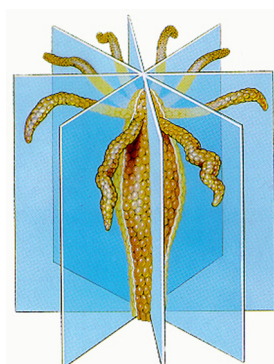
2.2.2 Symmetry

The regular arrangement of body parts in a geometrical design relative to the axis of the body is called symmetry. The animals, which can be cut into two equal parts, or antimeres in one or more planes passing through the '**principal axis**' of the body are called symmetrical animals. In a symmetrical animal, paired body parts are arranged on either side of the plane passing through the principal axis, such that they are equidistant from the plane. The unpaired body parts are located mostly on the plane, passing through the principal axis. Basically, the symmetry in animals is of two kinds:

- i) **Radial symmetry** and
- ii) **Bilateral symmetry**

i) **Radial Symmetry or Monaxial heteropolar (axis is single; poles are different) Symmetry**

When any plane passing through the central axis (**oro-aboral axis/ principal axis**) of the body divides an organism into two identical parts, it is called **radial symmetry**. The animals with radial symmetry are either sessile or planktonic or sluggish forms. It is the principal symmetry of the diploblastic animals such as the cnidarians and ctenophores (considered as biradial animals by some authors). Animals showing radial symmetry live in water and they can respond equally to stimuli that arrive from all directions. Thus, radial symmetry is an advantage to sessile or slow moving animals. However, triploblastic animals such as echinoderms are 'secondarily radially



Hydra- showing radial symmetry

symmetrical' (as it is five angled, it is also called **pentamerous radial symmetry**). Radially symmetrical animals have **many planes** of symmetry, whereas pentamerous radially symmetrical animals have **five planes** of symmetry.

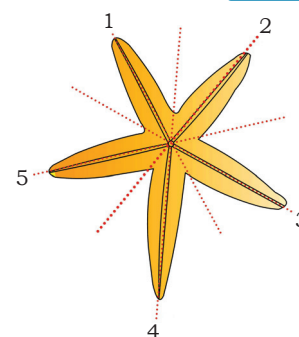
ii) Bilateral symmetry

When **only one plane** (*median sagittal plane*) that passes through the central axis (anterior – posterior axis) divides an organism into two identical parts, it is called **bilateral symmetry**. It is the '*principal type of symmetry*' in the **triploblastic** animals. Among the triploblastic animals, some **gastropods** become secondarily **asymmetrical**, though they have primarily bilaterally symmetrical larvae.

Symmetry of an organism is determined based on the symmetry of the early developmental stages and not that of the adults.

Bilaterally symmetrical animals are more efficient than the other animals in seeking food, locating mates and in avoiding or escaping from predators, because of **cephalization** (*concentration of nerve and sensory cells at the anterior end*). As a result of cephalization, bilaterally symmetrical animals can sense the new environment into which they enter and respond more efficiently and quickly.

Note: Sea anemones and acnidarians are said to show biradial symmetry.



Sea Star showing pentamerous symmetry

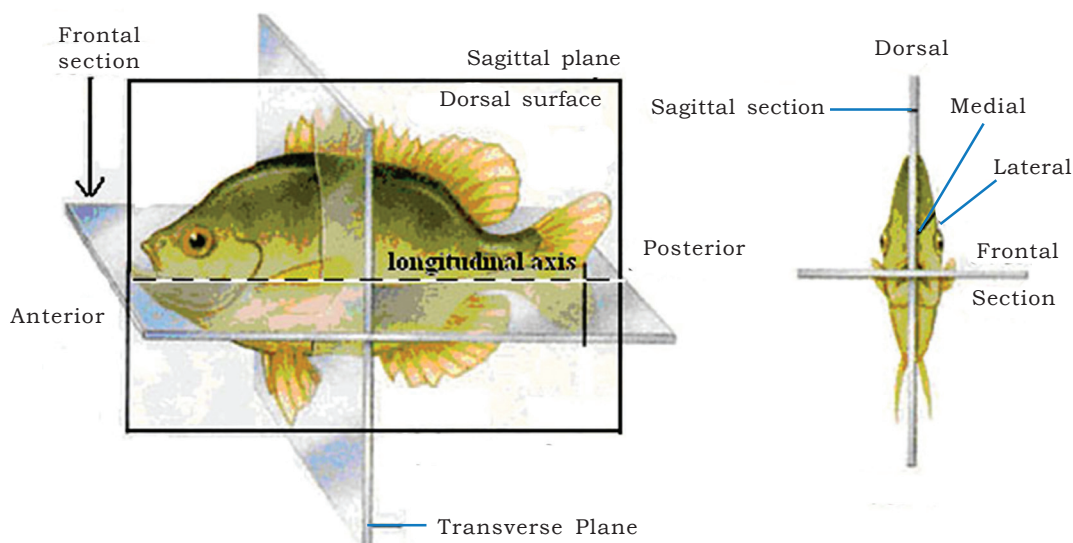


Fig. 2.4 Fish showing bilateral symmetry

2.3 COELOM

The term 'coelom' was coined by **Haeckel**. The body cavity, which is lined by mesoderm, is called coelom. More elaborately, coelom is a fluid-filled space between the body wall and visceral organs and lined by mesodermal epithelium, the peritoneum. Animals possessing coelom are called coelomates/eucoelomates. Evolution of efficient organ systems was not possible until the evolution of coelom for supporting the organs and distributing material.

2.3.1 Acoelomate bilaterians

The bilaterian animals in which the body cavity is absent are called acoelomates, e.g. Platyhelminthes (lowest bilaterians). In these animals, the mesenchyme derived from the third germinal layer, called mesoderm, occupies the entire blastocoel,

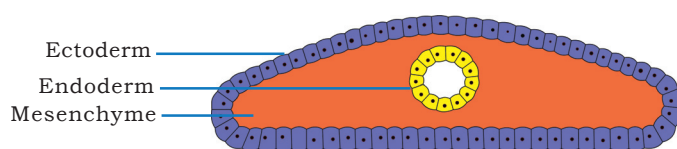


Fig. 2.5 T.S of acoelomate

between the ectoderm and the endoderm, so that the adults have neither the primary cavity (blastocoelom) nor the secondary cavity (coelom). As there is no body cavity, the acoelomates exhibit **solid body plan**. Problems faced by the

acoelomates due to absence of perivisceral cavity are - their internal organs cannot move freely, as they are embedded in the mesenchyme, diffusion of material from the gut to the body wall is slow and less efficient.

2.3.2 Pseudocoelomate bilaterians

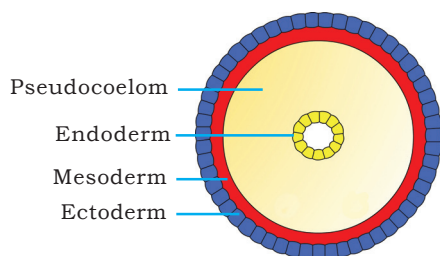


Fig. 2.6 T.S of pseudocoelomate

In some animals, the body cavity is not lined by mesodermal epithelia. Such animals are called Pseudocoelomates. They include the members of phylum Aschelminthes (Nematoda, Rotifera and some minor phyla). During the embryonic development, mesoderm (mesenchyme) occupies only a part of the blastocoel adjoining the ectoderm. The unoccupied portion of the blastocoel persists as **pseudocoelom**, which is filled with

pseudocoelomic fluid. Pseudocoelomates are the first animals to exhibit a '**tube-within-a-tube**' organisation.

Cnidarians and some flatworms have an incomplete gut and exhibit 'blind sac plan'. Justify the statement. Animals from nematodes to chordates in general have a complete gut and exhibit 'tube - within-a-tube plan'. Explain.

As the gut wall is made of only endodermal epithelium, diffusion of digested food from the lumen of the gut into the surrounding pseudocoelomic fluid becomes

easier and the absence of circulatory system is thus compensated. Though it is called pseudocoelom (false coelom), it performs almost all the functions of a regular coelom. Pseudocoelomic fluid of pseudocoelomates serves as a 'hydrostatic skeleton' and a 'shock absorber'. It allows the free movements of visceral organs, helps in the circulation of nutrients, and storage of nitrogenous wastes.

"Pseudocoelom is considered a false body cavity." Substantiate the statement?

2.3.3 Eucoelomate bilaterians

Coelom or 'true coelom' is a fluid-filled cavity, that lies between the body wall and the visceral organs and is lined by mesodermal epithelium, the peritoneum. The portion of the peritoneum that underlines the body wall is the **parietal peritoneum** or **somatic peritoneum**. The portion of the peritoneum that covers the visceral organs is the **splanchnic peritoneum** or **visceral peritoneum**. In coelomates, the visceral organs are suspended in the coelom by the peritoneum. A double layered peritoneum that connects some visceral organs to the body wall is called **mesentery**. In some eucoelomates such as the annelids, the dorsal and ventral mesenteries divide the coelom into paired compartments. Certain organs such as the kidneys of the vertebrates are covered by the parietal peritoneum only on their ventral side. Such a peritoneum is called the 'retroperitoneum' and the organs lined by it are called '**retroperitoneal organs**'.

During the embryonic development of the eucoelomates, the blastocoel is replaced by true coelom derived from the mesoderm. So, the true coelom is also called '**secondary body cavity**'. In the eucoelomates, mesoderm lines both the body wall and the walls of the visceral organs. So, visceral organs become 'muscular' and exhibit free movements 'independent' of the movement of the body wall in the coelomic fluid. As the wall of the gut becomes thick and muscular, digested food and other nutrients cannot easily diffuse from the lumen of the gut into the coelom i.e. fluid. Circulatory system (blood vascular system) is developed in the eucoelomates to overcome this problem. Based on the mode of formation of coelom, the eucoelomates are classified into two types:

I. Schizocoelomates

Animals in which the body cavity is formed by 'splitting of mesoderm' are called **schizocoelomates**. Annelids, arthropods and molluscs are schizocoelomates in the animal kingdom. All the schizocoelomates are **protostomians** and they show '**holoblastic**', '**spiral**' and '**determinate**' cleavage. **The 4d blastomere** or **mesentoblast cell** of the early embryo divides to form mesodermal blocks between the ectoderm and the endoderm and replaces the blastocoel. The split that appears in each

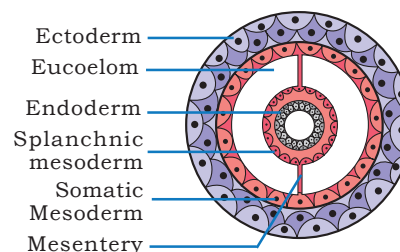


Fig. 2.7 T.S of eucoelomate

mesodermal block leads to the formation of '**schizocoelom**' (**split coelom**). In the annelids, the functional body cavity (perivisceral cavity) is schizocoelom. It is in the form of a series of paired coelomic cavities, but in arthropods and molluscs, the functional body cavity that lies around visceral organs is filled with blood (haemolymph) and is called **haemocoel**. It is formed by the fusion of the embryonic blastocoel with some true coelomic spaces and the tissues are directly bathed in the blood (haemolymph).

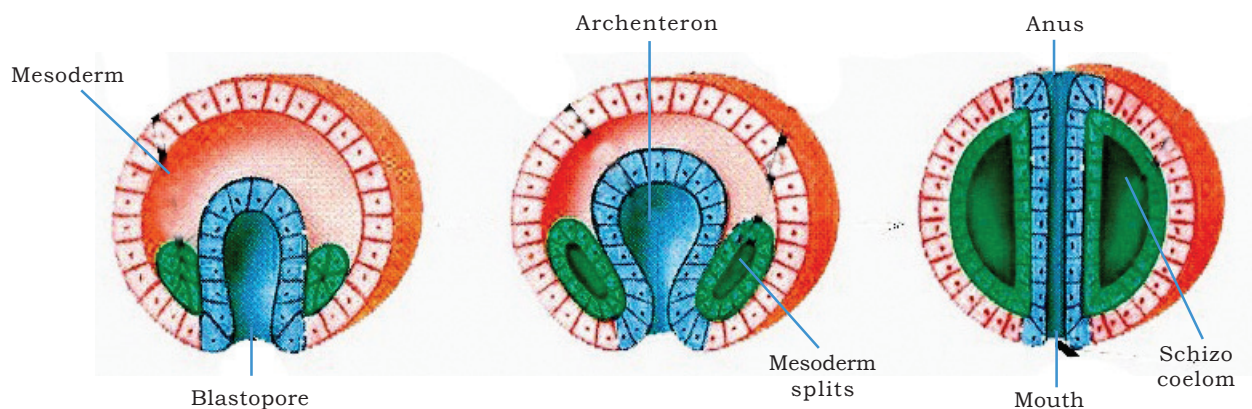


Fig. 2.8 Development of schizocoelom

II. Enterocoelomates

Animals in which the body cavity is formed from the mesodermal pouches of archenteron are called **enterocoelomates**. Echinoderms, hemichordates and chordates are the enterocoelomates. In these animals, mesodermal pouches that evaginate from the wall of the archenteron into the blastocoel are fused with one another to form the enterocoelom. All the enterocoelomates are **deuterostomes** and they show **radial** and **indeterminate** cleavage.

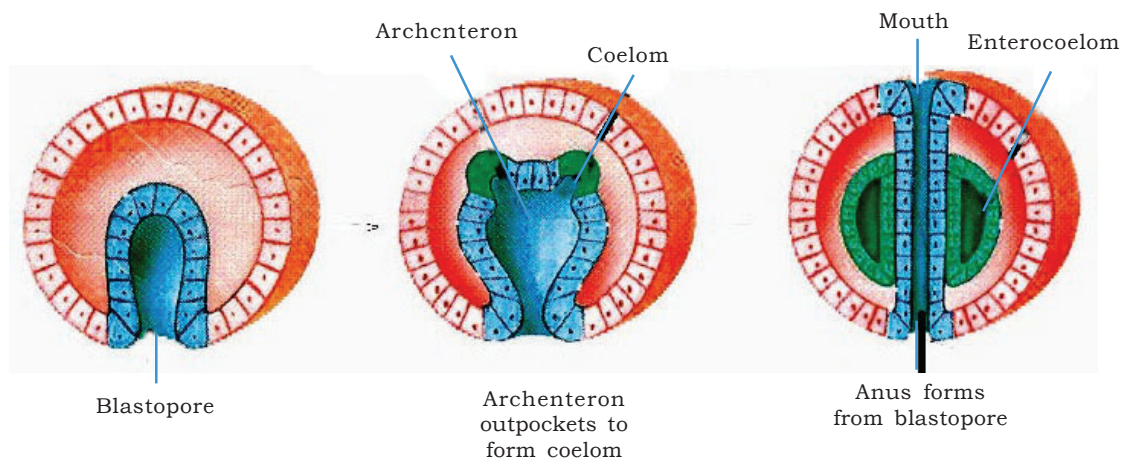
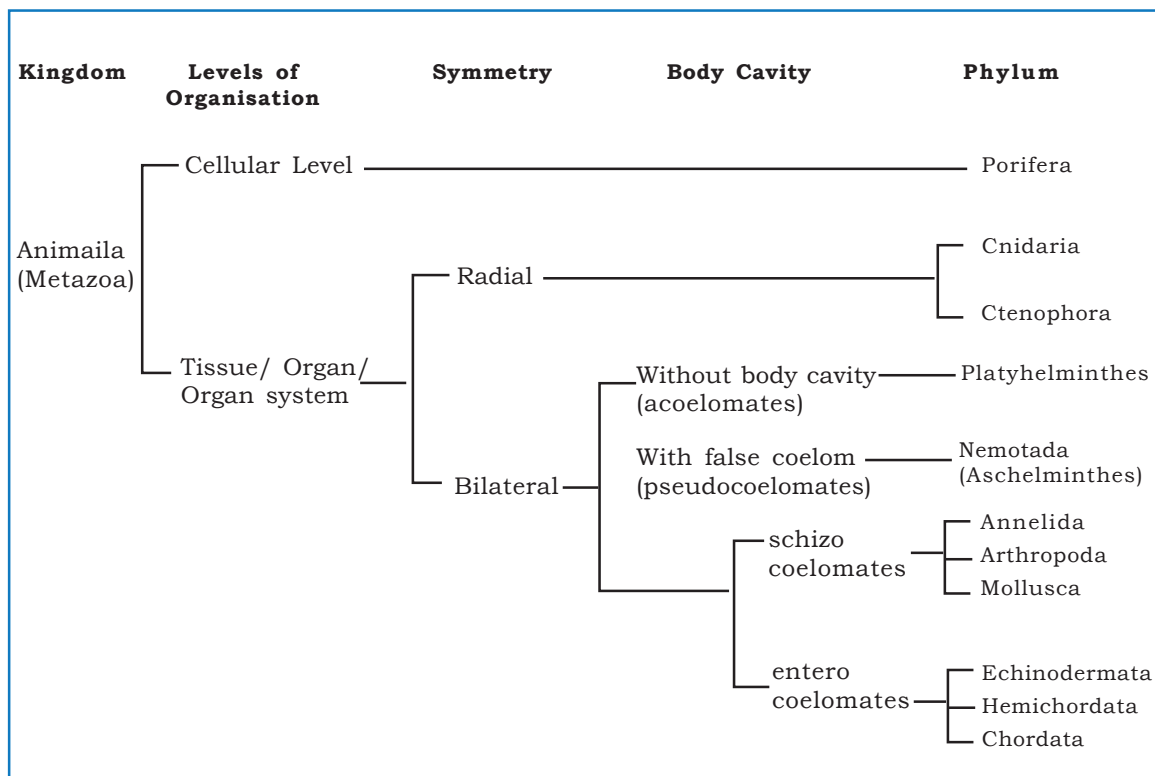


Fig. 2.9 Development of enterocoelom

Advantages of coelom over pseudocoelom

1. Visceral organs of eucoelomates are muscular (because of their association with mesoderm) and so they can contract and relax freely independent of the muscular movements of the body wall in the coelomic space, e.g. peristaltic movements of alimentary canal.
2. Gametes are released into the coelom in some invertebrates (which do not have gonoducts) and in the female vertebrates.
3. In non-chordates, coelomic fluid receives excretory products and stores them temporarily before their elimination.
4. In the eucoelomates, the mesoderm comes into contact with the endoderm of the alimentary canal, and it causes 'regional specialization of the gut, such as the development of gizzard, stomach etc. This is referred to as '**primary induction**'. In the case of the pseudocoelomates, due to the absence of such a contact between the gut and the mesoderm, the wall of the gut does not show complex and highly specialized organs.

The presence of a simple, thin walled gut in the nematodes allows easy diffusion of the digested foods into the pseudocoelomic fluid, thus compensating for the absence of circulatory system.



Summary of organisational levels, symmetry and body cavity

2.4 ANIMAL TISSUES

In acellular animals, all the functions like digestion, respiration, excretion and reproduction are performed by a single cell. In the complex body of multicellular animals, the same basic functions are carried out by different groups of cells in a well-organized manner. In the metazoans, a group of cells and cell products that arise from the same region of the embryo, which also perform a specific function, constitute a **tissue**. The study of tissues is known as **histology** or **microanatomy**.

Tissues are made up of cells and extracellular matrix. The extracellular matrix varies in composition and quantity from one type of tissue to the other. Animal tissues are broadly classified into four types: epithelial, connective, muscular and nervous tissues. These tissues differ from each other in types and functions of their cells, composition and relative amounts of the intercellular matrix.

2.4.1 Epithelial Tissues

Epithelium (epi,- upon; thelia: growing) forms the outer covering of the body and the lining of internal organs/cavities. Cells of an epithelium are held together by an intercellular '**cementing substance**'. The specialized 'junctions' provide both structural and functional links between the individual cells of an epithelium. They show different types of junctions so as to serve specific needs of that tissue. They are:

- A. Tight junctions:** These junctions between epithelial cells prevent 'leakages' of body fluids. For example, they prevent leakage of water into the surrounding cells in our sweat glands (making our skin water-tight). The plasma membranes of adjacent cells are tightly pressed against each other and are bound together by specific proteins.
- B. Desmosomes:** Epithelial and muscle cells are provided with '**desmosomes**' (**anchoring or adhesion junctions**) which act as 'rivets' binding the cells together into strong sheets. Intermediate filaments made of the protein '**keratin**,' anchor (support) the desmosomes in the cytoplasm.

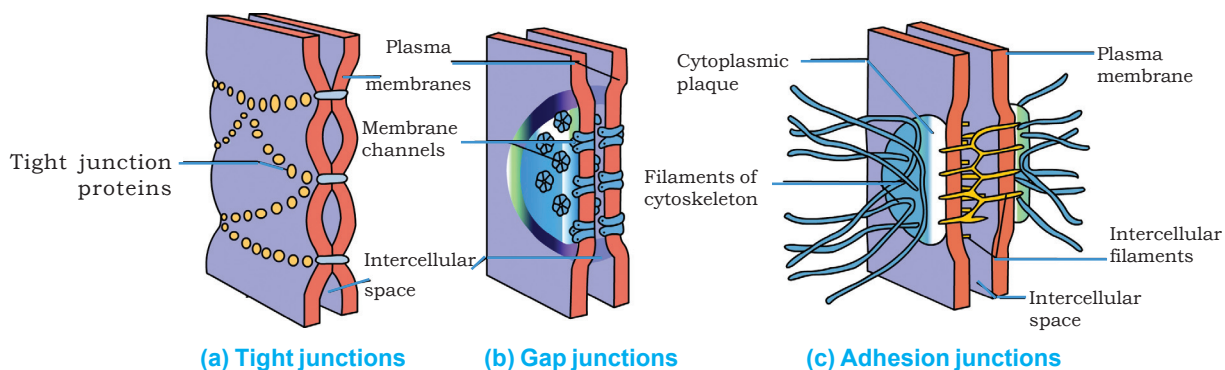


Fig. 2.10 Inter cellular junctions

C. Gap junctions (communicating junctions): They provide continuous ‘**cytoplasmic channels**’ between adjacent cells (comparable to the ‘**plasmodesmata**’ between adjacent plant cells). Various types of ions, sugar molecules, amino acids etc., can pass from a cell to an adjacent cell through ‘**gap junctions**’. They occur in many types of tissues including the ‘**cardiac muscles**’, where they allow rapid conduction of impulses or depolarisation.

An epithelium may be derived from any of the three germinal layers-ectoderm, endoderm or mesoderm. An epithelium rests on basement membrane, which is secreted partly by the epithelial cells and partly by the underlying tissue. Basement membrane consists of **basal lamina** that lies close to the epithelial cells and **reticular lamina** that lies close to the underlying connective tissue. Epithelial tissue is **avascular** and obtains nourishment from the underlying tissue by diffusion. Free surfaces of certain epithelial cells have microvilli or stereo cilia. **Microvilli** increase the surface area of absorptive cells. **Stereocilia** are long and non-motile cilia like processes seen in reproductive ducts (epididymis), internal ear etc. (See Glossary)

Stereocilia are non-motile, whereas cilia are motile. Why?

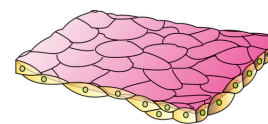
There are **two types of epithelial tissues** namely ‘simple epithelia’ and ‘compound epithelia’ based on the number of layers or strata. Various glands in the body involved in secretions are made up of epithelial tissue (glandular epithelium).

1. Simple epithelium

Simple epithelium is composed of a single layer of cells and forms the lining of body cavities, ducts and vessels. It helps in diffusion, absorption, filtration and secretion of substances. On the basis of the shape of the cells, it is further divided into three types:

i) Simple squamous epithelium (Pavement epithelium)

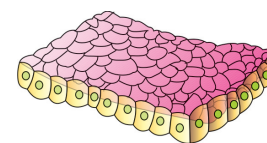
It is composed of a single layer of flat and **tile-like cells**, each with a centrally located ‘ovoid nucleus’. It is found in **endothelium** of blood vessels, **mesothelium** of body cavities (pleura, peritoneum, and pericardium), wall of **Bowman’s capsule** of nephron, lining of **alveoli** of lungs, etc.



Flattened cell

ii) Simple cuboidal epithelium

It is composed of a single layer of cube-like cells with centrally located spherical nuclei. It is found in the **germinal epithelium**, proximal and distal **convoluted tubules of nephron**. Cuboidal epithelium of proximal convoluted tubule of nephron has ‘**microvilli**’.



Cube-like cell

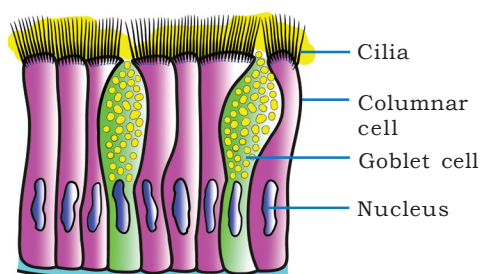


Fig. 2.11 Ciliated epithelium

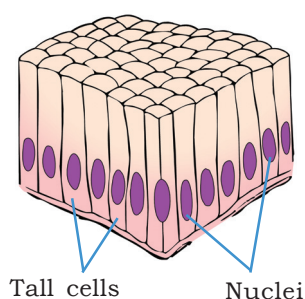
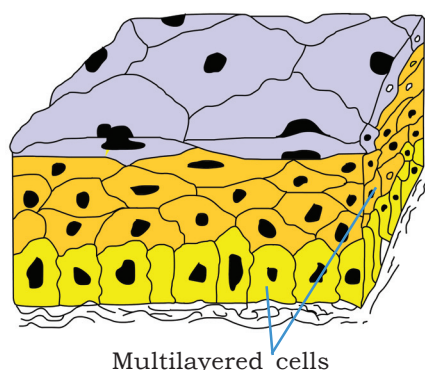


Fig. 2.12 Non - ciliated epithelium



iii) Simple columnar epithelium

It is composed of a single layer of tall and slender cells with oval nuclei located near the base. It has mucus-secreting 'goblet cells' in some places. It is of two types:

- a) **Ciliated columnar epithelium:** Columnar epithelial cells have cilia on their free surface. It is mainly present in the inner surface of hollow organs like **fallopian tubes**, **ventricles** of brain, **central canal** of spinal cord, **bronchioles**, etc.

Air passages, fluid filled cavities and genital ducts are usually lined with simple columnar ciliated epithelium. Why?

- b) **Non-ciliated columnar epithelium:** Columnar cells are without cilia. It is found in the lining of **stomach** and **intestine**. Microvilli are present in the columnar epithelium of intestine to increase the surface area of absorption.

2. Compound epithelium (stratified epithelium)

It is made up of more than one layer of cells. Its main function is to provide protection against chemical and mechanical stress.

- i. It covers the dry surfaces of the skin as **stratified, keratinized, squamous epithelium**.
- ii. It covers the moist surface of buccal cavity, pharynx, oesophagus and vagina as **stratified non-keratinized squamous epithelium**.
- iii. It forms the inner lining of the larger ducts of the salivary glands, ducts of sweat glands and pancreatic ducts as **stratified cuboidal epithelium**.
- iv. It forms the wall of the urinary bladder as **transitional epithelium**.

Pseudostratified epithelium. It is a simple epithelium, but gives the false appearance of stratified epithelium. It is due to the presence of nuclei at different levels and all cells are in contact with the basement membrane. It is either ciliated (as in trachea with goblet cells) or non-ciliated (as in epididymis without goblet cells).

Hair of man and horn of a *Rhinoceros* are homologous structures. Elucidate.

3. Glandular epithelium

Some of the columnar or cuboidal cells that get specialised for the production of certain secretions, form glandular epithelium. The glands are of two types -

i) Unicellular glands consisting of isolated glandular cells such as goblet cells of the gut.

ii) Multicellular glands, consisting of clusters of cells such as salivary glands.

On the basis of the mode of pouring (releasing) of their secretions, glands are divided into two types namely **exocrine** and **endocrine** glands.

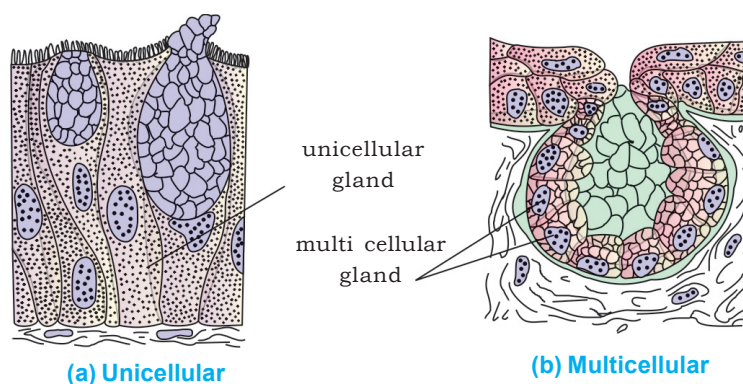


Fig. 2.13 Glandular epithelium

i. Exocrine glands are provided with ducts; secrete mucus, saliva, earwax (cerumen), oil, milk, digestive enzymes and other cell products.

ii. Endocrine glands are ductless and their products are '**hormones**', which are not sent out via ducts, but are carried to the target organs by blood.

Based on the mode of secretion, exocrine glands are further divided into

i. merocrine glands (e.g. pancreas) which release the secretory granules without the loss of other cellular material

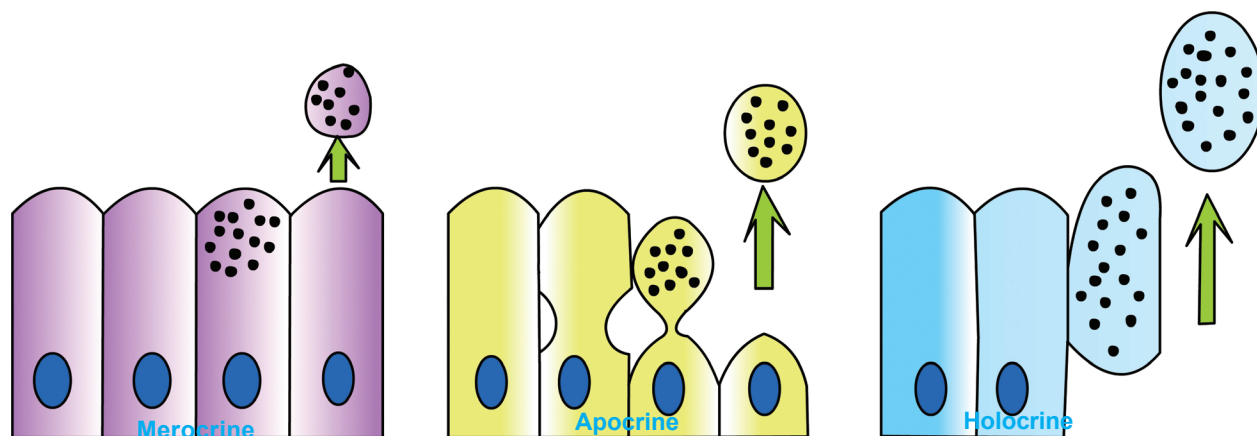


Fig. 2.14 Modes of secretion by exocrine glands

ii. Apocrine glands (e.g. mammary glands) in which the apical part of the cell is pinched off along with the secretory product and

iii. Holocrine glands (e.g. sebaceous glands), in which the entire cell disintegrates to discharge the contents.

2.4.2 Connective tissues

Connective tissues are the most abundant and widely distributed tissues in the body of higher animals. They are named 'connective tissues', because of their special function of binding and supporting other tissues or organs of the body. They are **mesodermal** in origin. Connective tissue consists of cells and large extracellular matrix. It consists of different combinations of fibres (collagen, elastic and reticular) and ground

substance which includes water, polysaccharides and proteins. The fibres provide strength, elasticity and flexibility to the tissue. Connective tissues are classified into three types namely connective tissue proper, skeletal tissue and fluid connective tissue based on the composition of matrix and types of cells.

I. Connective tissue proper

It is of two types,

1. **Loose connective tissue:** Cells and fibres are loosely arranged in a semi fluid ground substance. There are three types of loose connective tissues - areolar tissue, adipose tissue and reticular tissue.

i. Areolar tissue

It is one of the most widely distributed connective tissues in the body. It forms the packing tissue in almost all the organs. Areolar tissue forms subcutaneous layer of the skin. It has cells, fibres and empty spaces called 'areolae'. Cells of the areolar tissue are **fibroblasts**, **mast cells**, **macrophages**, **adipocytes** and **plasma cells**.

- 1) **Fibroblasts** are the most common cells which secrete fibres. The inactive cells are called **fibrocytes**.
- 2) **Mast cells** secrete **heparin** (an anticoagulant), **histamine**, **bradykinin** (vasodilators), and **serotonin** (vasoconstrictor). Vasodilators cause inflammation in response to injury and infection.
- 3) **Macrophages** are amoeboid cells, **phagocytic** in function and act as internal scavengers. They are derived from the monocytes of blood. 'Tissue fixed macrophages' are called **histiocytes** and others are 'wandering macrophages'.
- 4) **Plasma cells** are derived from the B-lymphocytes and produce **antibodies**.
- 5) **Adipocytes** are specialized cells for the storage of fats.

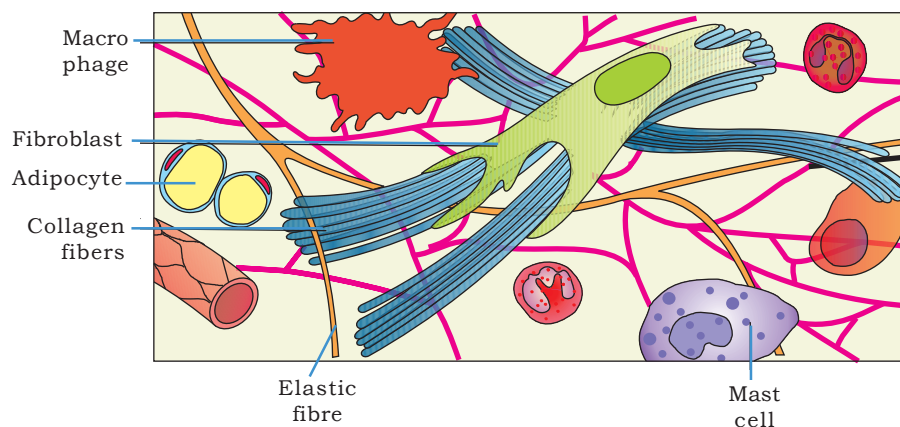


Fig. 2.15 (a) Areolar tissue

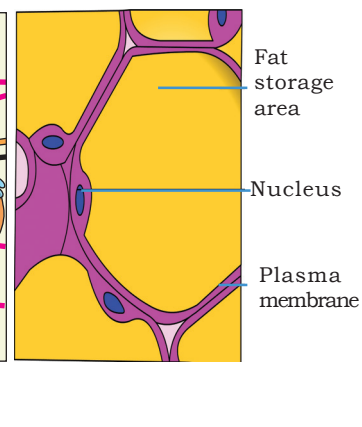


Fig. 2.15 (b) Adipose tissue

Fibres of areolar tissue are of three types. They are collagen, reticular and elastic fibres. Collagen and reticular fibres are composed of the protein **collagen**, elastic fibres are made of the protein **elastin**. **Collagen (white) fibres** occur in bundles, and they are **strong** and **stretch resistant**. **Reticular fibres** are thin and form a network and they provide **strength** and **support** to certain tissues such as bone marrow. **Elastic (yellow) fibres** are branched. They are also found in elastic cartilages, elastic ligaments, etc.

ii. Adipose tissue

It is specialized for **fat storage**. It consists of a large number of **adipocytes** (fat storing cells) and few fibres. Adipose tissue which is found beneath the skin provides **thermal insulation** e.g., **blubber** of aquatic mammals such as whales and sea cows. It acts as a storage tissue in the **hump** of camel. It acts as **shock absorber** in palms and soles. Adipose tissue is of two types: white adipose tissue, brown adipose tissue. Excess nutrients which are not used immediately are converted into fats and stored in this tissue.

White adipose tissue (WAT): It is the predominant type in the adults, and the adipocyte has a single large lipid droplet (monolocular). White fat is metabolically less active.

Brown adipose tissue (BAT): It is found in foetuses and infants. Adipocyte of BAT has several small 'lipid droplets' (multilocular) and numerous mitochondria. Brown fat is metabolically active and generates 'heat' to maintain body temperature required by infants. It is comparatively more vascular, hence the brown colour.

iii. Reticular tissue

It has specialized fibroblasts called **reticular cells**. They secrete 'reticular fibers' that form an inter connecting network. It forms the 'supporting frame work' of **lymphoid organs** such as bone marrow, spleen and lymph nodes and forms the **reticular lamina** of the 'basement membrane'.

2. Dense connective tissue

This tissue consists of more fibres, but fewer cells. It has very little ground substance. Based on the arrangement of fibres, dense connective tissue is of three types.

i. Dense regular connective tissue

In this tissue, collagen fibres are arranged parallel to one another in bundles. **Tendons** which attach the skeletal muscles to bones and **ligaments** which attach bones to other bones are examples of this type of connective tissue.

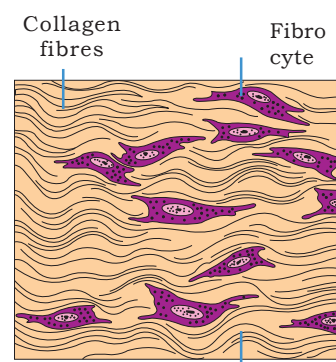


Fig. 2.16 Dense regular fibrous tissue

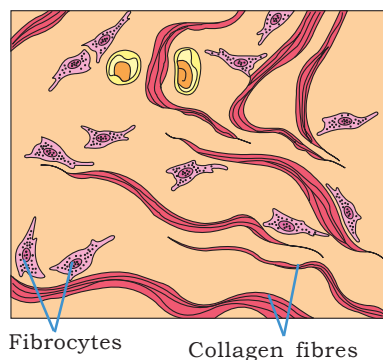


Fig. 2.17 Dense irregular fibrous tissue

ii. **Dense irregular connective tissue** : In this type of connective tissue, bundles of collagen fibres are irregularly arranged. Periosteum, endosteum, pericardium, heart valves, joint capsule and deeper region of dermis of skin contain / are made up of this type of connective tissue.

iii. **Elastic connective tissue**: It is mainly made of yellow elastic fibres, capable of considerable extension and recoil. This tissue can recoil to its original shape, when the forces of stretch are released. It occurs in the walls of arteries, **vocal cords, trachea, bronchi** and '**elastic ligaments**' present between vertebrae.

In addition to the above mentioned connective tissues, **mucous connective tissue** occurs as foetal or embryonic connective tissue. It is present in the umbilical cord as **Wharton's jelly**.

II. Skeletal tissues / supporting tissues

It forms the endoskeleton of the vertebrates. It supports the body, protects various organs, provides surface for the attachment of muscles and helps in locomotion. It is of two types:

1. Cartilage (Gristle)

Cartilage is a solid, but **semi-rigid (flexible) connective tissue**. It resists compression. Matrix is firm, but somewhat pliable. It has collagen fibres, elastic fibres (only in the elastic cartilage) and matrix-secreting cells called **chondroblasts**. These cells are enclosed in fluid filled spaces called **lacunae**. Chondrocytes are the inactive cells of a cartilage. Cartilage is surrounded by a fibrous connective tissue sheath called **perichondrium**. Cartilage is '**avascular**' and it is nourished by 'diffusion of nutrients' from the blood capillaries of the perichondrium. Growth, regeneration and repair of cartilage take place by the activity of **perichondrial** cells. Cartilage is of three types, which differ from each other chiefly in the composition of the matrix.

- i. **Hyaline cartilage** : It is bluish-white, translucent and glass-like cartilage. Matrix is homogeneous and shows delicate **collagen fibres**. It is the **weakest** and the **most common** type of all the cartilages. Perichondrium is present except in **articular cartilages**. It forms the embryonic endoskeleton of bony vertebrates, endoskeleton of cyclostomes (jawless vertebrates) and adult cartilaginous fishes. It forms the **articular cartilages** (free surfaces of long bones that form joints), **costal cartilages** (sternal parts of ribs), and the **epiphyseal plates**. It also forms the **nasal septal cartilage, cartilaginous rings** of trachea, bronchi and **cartilages of larynx**.

Degeneration of which part of a long bone results in arthritis (a disease which limits the movements of bones causing severe pain in the joints).

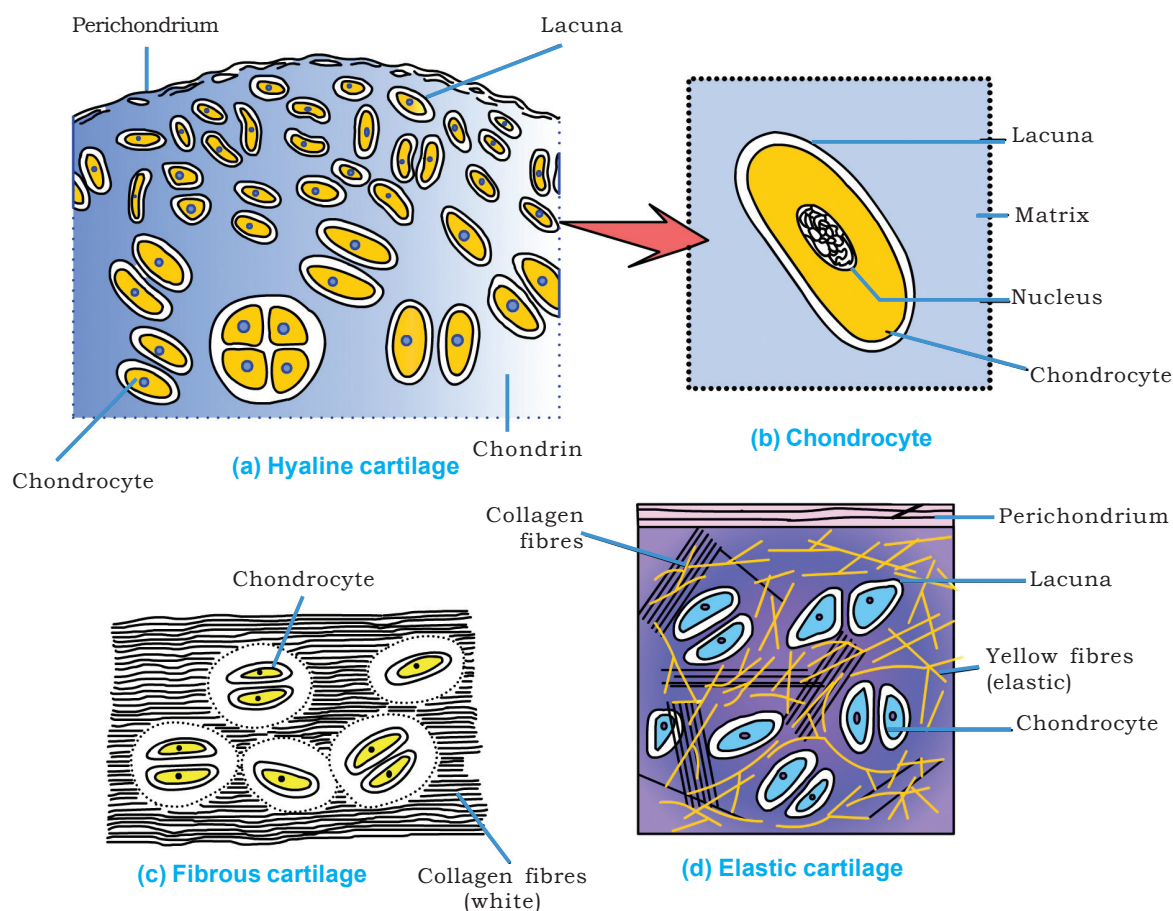


Fig. 2.18 Types of cartilage

ii. Elastic cartilage: It is yellowish due to elastic fibres. Matrix has abundance of yellow elastic fibres in addition to collagen fibres. It provides strength and elasticity. Perichondrium is present. It is found in the **pinnae of the external ears, Eustachian tubes and epiglottis**.

iii. Fibrous cartilage: Matrix has bundles of collagen fibres. **Perichondrium** is absent. It is **the strongest** of all the types of **cartilages**. It occurs in the **inter-vertebral discs** and **pubic symphysis** of the pelvis.

2. Bone (osseous) tissue

Bone is highly calcified (mineralized), solid, hard and rigid connective tissue. It is the major component of the endoskeleton of most adult vertebrates. It is the main tissue that provides structural framework to the body. It supports the soft tissues, and protects the delicate organs. Limb bones of animals serve the weight-bearing functions. Bones also interact with muscles attached to them to bring about movements. Bones have a hard and non-pliable matrix, rich in **calcium salts** and **collagen fibres** which give the bone its strength. During ageing, the proportion of inorganic materials increases in a

bone, making it more brittle. Bone forms **homeostatic reservoir** of calcium, magnesium, phosphorus, etc. Bone is **highly vascular**.

Bone has an outer fibrous connective tissue sheath called **periosteum**, the inner connective tissue sheath that lines the marrow cavity called **endosteum**, non-living extra cellular matrix, living cells and bone marrow. Bone cells include **osteoblasts**, **osteocytes** and **osteoclasts**. **Osteoblasts** (immature bone cells) secrete the organic components (collagen fibres) of matrix and also play an important role in 'mineralization of bone' and become **osteocytes** (mature bone cells). Osteocytes are enclosed in fluid filled lacunae. **Osteoclasts** are phagocytic cells involved in resorption of bone (demineralisation).

A. Types of bones:

a. Types of bones based on the method of formation

- i) **Cartilage bones** (replacing bones or **endochondral bones**) are formed by ossification within the cartilage e.g. bones of limbs, girdles and vertebrae.
- ii) **Investing bones** (**membrane bones** or **dermal bones**) are formed by the ossification in the embryonic mesenchyme e.g. most of the bones of **cranium**.
- iii) **Sesamoid bones** are formed by ossification in tendons e.g. **patella** (knee cap) and **pisiform** bone of the wrist of a mammal.
- iv) **Visceral bones** are formed by ossification in the soft tissues, e.g. **Os cordis** (inside the heart of ruminants), **Os penis** (inside the glans-penis of many mammals such as the rodents, bats and carnivores).

b. Types of bones based on the structure

- i. Spongy bone (Cancellous bone or trabecular bone) : It occurs in the epiphyses and metaphyses of long bones. It looks spongy and contains columns of bone called '**trabeculae**' with irregular interspaces filled with red bone marrow.
- ii. Compact bone: The diaphysis of a long bone is made up of 'compact bone'. It has dense continuous lamellar matrix between periosteum and endosteum.

B. Structure of a compact bone

Diaphysis (shaft) is a part of a long bone that lies in between expanded ends (*epiphyses*). In a growing bone there is a region called metaphysis between the diaphysis and epiphysis. It consists of an epiphyseal plate (formed by hyaline cartilage). It helps in the elongation of the bone. In adults it is represented by a **bony epiphyseal line**. Diaphysis is covered by a dense connective fibrous tissue called **periosteum**. Diaphysis of a long bone has a hollow cavity called **marrow cavity** which is lined or surrounded by the **endosteum**. In between periosteum and endosteum, the matrix of the bone is laid down in the form of 'lamellae'. **Outer circumferential lamellae** are located immediately beneath the periosteum; **inner**

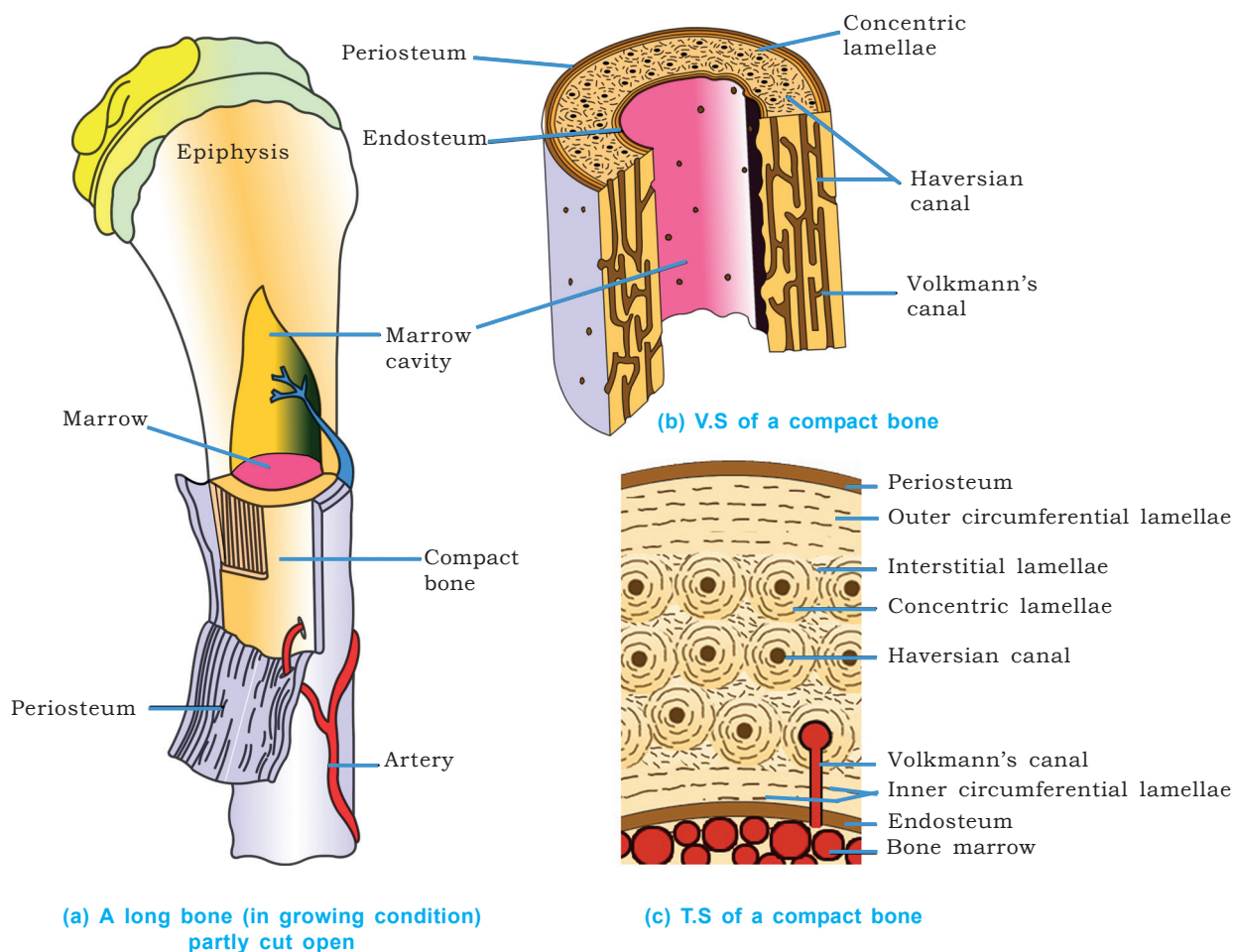


Fig. 2.19 Osseous tissue

circumferential lamellae are located around the endosteum. Between the outer and inner circumferential lamellae, there are many Haversian systems (osteons - units of bone). The spaces between the Haversian systems are filled with **interstitial lamellae**. **Haversian system** consists of a Haversian canal that runs parallel to the marrow cavity. It contains an artery, a vein and a lymphatic vessel. Haversian canal is surrounded by **concentric lamellae**. Small fluid filled spaces called 'lacunae' provided with minute **canaliculi** lie in between the lamellae. Canaliculi connect the lacunae with one another and with Haversian canal. Each lacuna encloses one osteocyte (inactive form of osteoblast). The cytoplasmic processes of osteocytes extend through canaliculi and get inter connected into a network of cells. A Haversian canal and the surrounding lamellae and lacunae are collectively called a **Haversian system** or **osteon**. The Haversian canals communicate with one another, with the periosteum and also with the marrow cavity by transverse or oblique canals called **Volkman's canals**. Nutrients and gases diffuse from the vascular supply of Haversian canals.

Of the dry weight of a fully grown bone, inorganic matter constitutes about 65% and organic matter constitutes about 35%. Major part of inorganic matter is calcium phosphate which is present in the form of **crystals** of **hydroxyapatite**. The major organic substance is **collagen**.

Note : When bones are broken, the osteocytes are reactivated to osteoblasts and start deposition of mineral material, thus joining the fractured bones.

If, a bone is kept in dilute acid (HCl) it becomes soft. Can you explain why?

III. Fluid connective tissue (Vascular tissue): It is a tissue, which consists of fibre-free 'liquid matrix' called plasma and living cells that do not produce the matrix. It includes blood and lymph.

1. Blood

It is a red coloured, opaque and slightly alkaline fluid. The study of blood is called **haematology**. It is a tissue that circulates to various parts of the body through **cardiovascular system**. It is composed of blood plasma and 'formed elements' or blood cells, the RBC, WBC and platelets. The plasma constitutes about 55% and the cells about 45% of the total volume of blood. The total volume of blood in an adult human being is about 5 to 6 litres. The percentage of total volume occupied by RBCs is called **haematocrit value** or '**packed cell volume**'.

A. Plasma

It is the fluid matrix of blood. It consists of 92% of water and 8% of solutes. Solutes include plasma proteins, mineral salts, nutrients, gases, excretory wastes, enzymes, hormones, etc. Plasma proteins include albumins, globulins and blood clotting proteins such as **fibrinogen**, **prothrombin** and anticoagulant heparin all of which are produced by the liver. Globulins are formed in the liver and lymphoid organs. **Albumin** is the smallest and the most abundant serum protein and it is mainly responsible for '**colloidal osmotic pressure**' of blood. Fall in the levels of albumin in blood plasma, results in **oedema** (accumulation of fluids in tissues). Globulins are of three types, alpha, beta and gamma. Gamma globulins are the antibodies, also called immunoglobulins. Plasma proteins act as '**acid-base buffers**', maintaining the pH of blood at 7.4. Salts of blood plasma are the chlorides, bicarbonates, sulphates, phosphates of Na, K, Ca, and Mg.

B. Formed elements

They include **erythrocytes** (red blood corpuscles), **leucocytes** (white blood corpuscles) and **platelets**. The process of formation of blood cells is called **haemopoiesis** or **haematopoiesis**. In the earliest stages of embryogenesis, blood cells are formed from the **yolk sac mesoderm**. Later on, the **liver** and the **spleen** serve as "temporary haemopoietic tissues". In the final stage of embryonic development and after birth, the **red bone marrow** is the primary site of haemopoiesis.

i. Red blood corpuscles (Erythrocytes)

Erythrocytes of mammals are circular (elliptical in camels and Llamas), biconcave and enucleate. The biconcave shape provides a large surface area-to-volume ratio, thus providing more area for the exchange of gases. These are $7.8\text{ }\mu\text{m}$ in diameter. The number of RBC per cubic millimeter of blood is about 5 million in a man, and 4.5 million in a woman. Decrease in the number of erythrocytes is called **erythrocytopenia** and it leads to **anaemia**. An abnormal rise in RBC count is called **polycythemia**. Shortage of oxygen stimulates the kidneys to secrete a hormone called **erythropoietin** into the blood. Erythropoietin stimulates the bone marrow to increase the production of RBC. **Vitamin B₁₂** and **folic acid** are required for maturation of RBC. RBCs arranged in the form of rolls of coins called '**rouleaux**'.



RBC

Mammalian RBC is surrounded by plasma membrane. Nucleus and other cell organelles are lost in the **reticulocyte** stage of its development. Cytoplasm of RBC contains a chromo protein, the 'haemoglobin'. Each Haemoglobin molecule consists of **4 polypeptide (2 α & 2 β)** chains and **4 haeme** molecules. In the centre of each haeme group is **one Fe²⁺**, which can combine with one molecule of O₂. Life span of RBC in humans is about **120 days**. The worn out RBC are destroyed in the '**spleen**' and '**liver**'.

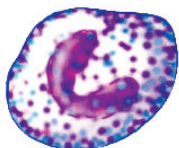
RBC of mammals are biconcave and enucleate. What is the advantage of it?

ii. White blood corpuscles (Leucocytes)

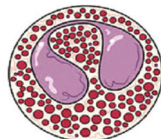
These are nucleate, colourless, complete cells. They are spherical or irregular in shape, and are capable of exhibiting amoeboid movement into the extravascular areas by **diapedesis**. They are larger than RBC in size, and less than RBC in number. The total leucocyte count is 6,000 - 10,000 per cubic millimeter of blood under normal conditions. The process of formation of WBC is called **Leucopoiesis**. Slight increase in the WBC count is called **Leucocytosis** (during infection and allergy). An abnormal increase in the number of WBC is indicated in a type of cancer called **Leukemia**. Fall in WBC count is called **Leucocytopenia**. **WBC** are of two main types: **Granulocytes** and **Agranulocytes**.

Granulocytes

They possess cytoplasmic granules (specific granules) that may take one of the three different types of stains, neutral or acidic or basic. Nucleus of the granulocytes is divided into lobes and assumes different shapes, hence, these are also called **Polymorph-nuclear leucocytes**. Based on the staining properties these are of three types:



Basophil



Eosinophil



Neutrophil

Basophils

They are the least common type of WBC and constitute about 0.4% of the total leucocytes. Nucleus is partially divided into 2 or 3 lobes. Cytoplasmic granules are 'fewer' and 'irregular' in shape. They take basic stains. They produce **heparin**, **histamine**, etc. They supplement the function of mast cells when needed.

Eosinophils (acidophils)

They constitute about 2.3% of the total leucocytes. Nucleus is distinctly **bilobed**. Cytoplasm has large granules which stain with acidic dyes such as '**eosin**'. They play a role in **allergic reactions**. Their number increases during 'allergic reactions' and 'helminth infections'. They remove '**antigen - antibody complexes**', from the blood. They are capable of phagocytosis and are not as effective as neutrophils in killing bacteria.

Neutrophils

They are the most common type of WBC and constitute about 62% of the total leucocytes. Nucleus is **many lobed** (2-5). Specific cytoplasmic granules are small and abundant. They stain with 'neutral dyes'. These are active **phagocytic cells** commonly described as '**microscopic policemen**'. Certain neutrophils of female mammals have **sex chromatin body** or **Drumstick body** (an extra 'X' chromosome) attached to the nucleus.

Agranulocytes

Cytoplasmic granules are absent in agranulocytes. Nucleus of these cells is not divided into lobes. These are of two types:



Lymphocyte

Lymphocytes: They constitute about 30% of the total leucocytes. They are small, spherical cells with **large spherical nucleus** and scanty peripheral cytoplasm. There are functionally two types of lymphocytes - 'B' lymphocytes, which produce 'antibodies' and 'T' lymphocytes which also have an important role in the immunological reactions of the body.

Some lymphocytes live for only a few days while others survive many years.

Monocytes

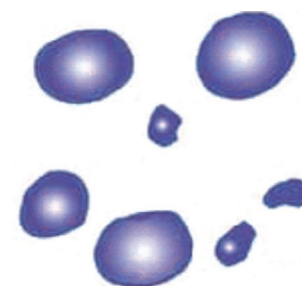
They constitute about 5.3% of the leucocytes. The nucleus is **kidney shaped** (reniform). These are the largest, motile phagocytes. They engulf bacteria and cellular debris (wastes). They differentiate into **macrophages**, when they enter the connective tissues.



Monocyte

iii. Blood platelets (Thrombocytes)

These are colourless non-nucleated, round or oval biconvex discs. Number of platelets per cubic mm of blood is about 2,50,000 - 4,50,000. They are formed from giant **megakaryocytes** produced in the red bone marrow, by **fragmentation**. The average life-span of blood platelets is about 5 to 9 days. They secrete **thromboplastin** and play an important role in blood clotting. They adhere to the damaged endothelial lining of capillaries and 'seal' minor vascular openings.



Blood platelets

2. Lymph

Lymph is a colourless fluid. It lacks RBC, platelets and large plasma proteins, but has more number of leucocytes. It is chiefly composed of plasma and lymphocytes. When compared to the tissue fluid, it contains very small amounts of nutrients (except fats) and oxygen, but has abundant CO_2 and other metabolites. The most important site of formation of lymph is **interstitial space**. As blood passes through the blood capillaries, some portion of blood that includes water, solutes and proteins of low molecular weight passes through the walls of capillaries, into the interstitial spaces due to hydrostatic pressure at the arteriolar ends. This fluid forms the **interstitial fluid** (tissue fluid). Most of the interstitial fluid is returned directly to the capillaries due to osmotic pressure at the venular ends. Little amount of this tissue fluid passes through a system of lymphatic capillaries (lymph capillaries of the intestinal villi are called 'lacteals'), vessels, lymph ducts and finally reach the blood through the subclavian veins. The extracellular 'tissue fluid' after passing into the lymph capillaries and lymph vessels is called 'lymph'. Lymphatic system represents an '**accessory route**' by which interstitial fluid flows from tissue spaces into blood.

2.4.3 Muscular Tissue

Muscular tissue is **mesodermal** in origin, except muscles of the **iris** and **ciliary body**, which are **ectodermal** in origin. Muscles show three essential properties such as **excitability**, **conductivity** and **contractility**. Muscle fibers contract (shorten) in response to stimulation, then relax (lengthen or return to their un-contracted state), in a coordinated fashion. Muscles play an active role in the movements of the body to adjust to changes in the surrounding environment and to maintain the posture of the body. The study of muscular tissues is known as **myology**. Muscular tissue has

elongated cells called 'muscle fibers' (myocytes) which are surrounded by a connective tissue sheath. Extracellular matrix is absent. The plasma membrane of a muscle fibre is called **sarcolemma**. The cytoplasm of a muscle fibre is called **sarcoplasm**, the

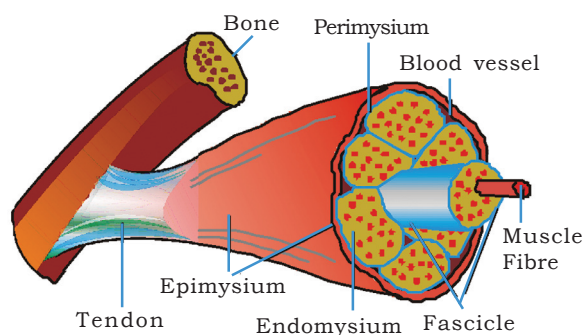


Fig. 2.20 (a) Skeletal muscle

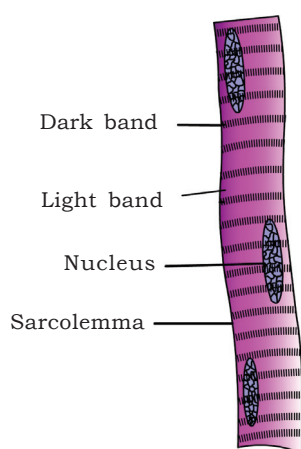


Fig. 2.20 (b) Striped muscle fibre

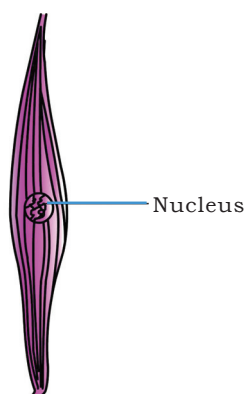


Fig. 2.20 (c) Smooth muscle fibre

endoplasmic reticulum, the **sarcoplasmic reticulum**, and the mitochondria, the **sarcosomes**. The cytoplasm of a muscle fibre has several **myofibrils**. Each myofibril has thick (**myosin**) and thin (**actin**) **myofilaments**. The regular arrangement of myosin and actin filaments is responsible for the alternate dark and light bands of a 'striated muscle'. Sarcoplasm also contains ATP, phosphocreatine, glycogen and myoglobin. Muscles are of three types - **skeletal**, **smooth** and **cardiac**.

1. Skeletal (striped and voluntary) muscle

It is usually attached to skeletal structures by 'tendons'. In a typical muscle such as the 'biceps' muscle, skeletal muscle fibre is surrounded by a thin connective tissue sheath, the **endomysium**. A bundle of muscle fibres is called a **fascicle**. It is surrounded by a connective tissue sheath called **perimysium**. A group of fascicles form a 'muscle' which is surrounded by an **epimysium** (outer most connective tissue sheath). These connective tissue layers may extend beyond the muscle to form a chord-like **tendon** or sheet-like **aponeurosis**.

A skeletal muscle fibre is a **long, cylindrical** and **unbranched cell**. It is a **multinucleated** cell with many oval nuclei characteristically in the "peripheral" cytoplasm (a syncytium formed by fusion of cells). Sarcoplasm has many **myofibrils** which show alternate dark and light bands. So it is called **striped** or **striated** muscle. Skeletal muscle usually works under the conscious control of an organism (a **voluntary muscle**). Skeletal muscle **contracts quickly** and undergoes **fatigue quickly**. They are innervated by the 'somatic nervous system'. **Satellite cells** are quiescent (quiet and inactive), mononucleate and myogenic cells (cells that can generate muscle cells) and help in regeneration, which is 'limited'.

2. Smooth (unstriated and involuntary) muscle

It is located in the walls of the **visceral organs** such as blood vessels, trachea, bronchi, stomach, intestine, excretory and genital

ducts, and so this is also called '**visceral muscle**'. As cross striations are absent, it is called '**smooth muscle**'. It is also found in **iris** and **ciliary body** of eye and in the skin as '**arrector pili muscles**' that are attached to hair follicles.

Usually smooth muscles are arranged in 'layers'/'sheets'. A smooth muscle fibre is a spindle shaped (fusiform), **uninucleate** cell. Myofibrils do not show alternate dark and light bands due to irregular arrangement of actin and myosin molecules. They do not work under the conscious control, and so they are called **involuntary muscles**. Smooth muscles exhibit '**slow**' and '**prolonged**' contractions. They may remain contracted for long periods **without fatigue** (show sustained involuntary contractions called '**spasms**'). The contraction of smooth muscles is under the control of the **autonomous nervous system**.

3. Cardiac (striped and involuntary) muscle

The cardiac muscle is striated like the skeletal muscle (shows **sarcomeres**). Cardiac muscle is found in the 'myocardium' of the heart of vertebrates.

The cardiac muscle cells or the 'myocardial cells' **are short, cylindrical, mononucleate or binucleate cells** whose ends branch and form junctions with other cardiac muscle cells. Each myocardial cell is joined to adjacent myocardial cells by 'electrical synapses' or '**gap junctions**'. They permit 'electrical impulses' to be conducted along the long axis of the cardiac muscle fibre. The dark lines across cardiac muscle are called **intercalated discs (IDs)**. These discs are characteristic of the cardiac muscle and are called communication junctions which allow the cells to contract as a unit. Intercalated discs contain three different types of cell junctions and the 'gap junctions' which are responsible for rapid conduction of action potentials.

The cardiac muscle of vertebrates does not require any nerve stimulus to contract because it can produce impulses spontaneously from a specialised auto-rhythmic structure, the '**pace maker**' (myogenic heart). As such, the cardiac muscle is **involuntary** in its function. However, the rate of heart beat is regulated by an autonomic innervation and hormones such as **epinephrine/adrenaline**. The excitation of one myocardial cell results in the excitation of all other myocardial cells quickly to produce a '**whole hearted contraction**', of the entire muscle as a single unit. Thus the cardiac muscle is described as a '**functional syncytium**'. The cardiac muscle is **highly resistant to fatigue**, because it has numerous sarcosomes, many molecules of myoglobin (oxygen storing pigment) and rich supply of blood which facilitate '**continuous aerobic respiration**'.

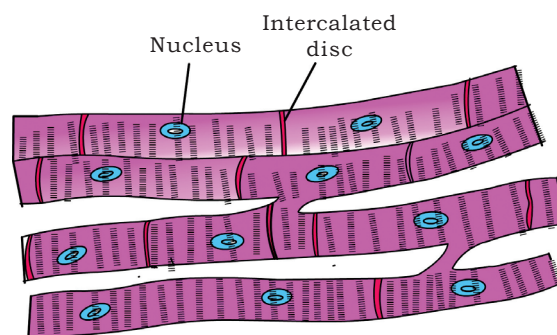


Fig. 2.20 (d) Cardiac muscle

2.4.4 Nervous Tissue

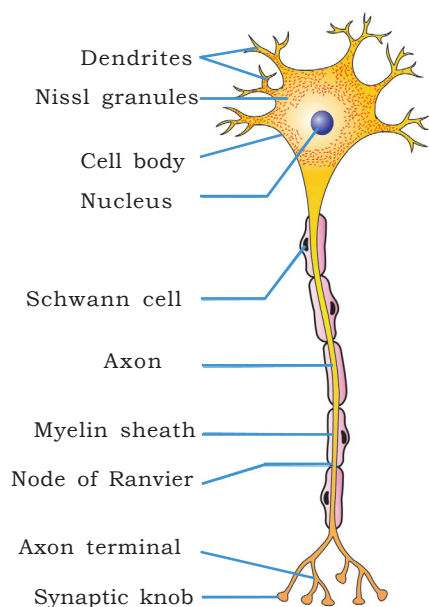


Fig . 2.21 Multipolar Neuron (Structure)

Nervous tissue is derived from the **ectoderm**. It exerts the greatest control over the body's responsiveness to the changing external and internal environment. It consists of two types of cells. The **neurons** (impulse conducting cells) and the **neuroglia** (non-conducting cells or **supporting cells**)

I. Neurons

Neurons are the 'functional units' of nervous tissue. These are electrically excitable cells which receive, initiate, and conduct/transmit impulses. When a neuron is stimulated, an electric disturbance (**action potential**) is generated due to the formation of a series of action potentials, which swiftly travels along its plasma membrane. A neuron usually consists of a "**cell body**" with one to many dendrites and a single axon.

1. Cell body

It is also called **perikaryon**, **cyton** or **soma**. It contains abundant granular cytoplasm and a large spherical nucleus. The cytoplasm has **Nissl bodies** (they represent **RER**, the sites of protein synthesis), neurofibrils and **lipofuscin** granules (the products of cellular wear and tear, accumulating in lysosomes with age). A group of cell bodies in the central nervous system is called a '**nucleus**', and in the peripheral nervous system, it is called a '**ganglion**'.

2. Dendrites

Several short, branched processes which arise from the cyton are called dendrites. They also contain **Nissl bodies** and **neurofibrils**. They conduct nerve impulses towards the cell body (**afferent processes**).

3. Axon

An axon is a single, long, cylindrical process that originates from a region of the cyton called **axon hillock**. Plasmalemma of an axon is called **axolemma**, and the cytoplasm is called **axoplasm**, which contains neurofibrils. However, Nissl bodies are absent. An axon may give rise to **collateral branches**. Distally it branches into many fine filaments called **telodendria** (**axon terminals**), which end in bulb like structures called **synaptic knobs** or **terminal boutons**. Synaptic knobs possess 'synaptic vesicles' containing chemicals called **neurotransmitters**. Axon transmits nerve impulse away from the cyton (**efferent process**) to an interneuronal or neuro-muscular junction called synapse. Groups of axons (nerve fibres) in the central nervous system (CNS) are called **tracts** and in the peripheral nervous system (PNS), they are called **nerves**.

What type of neurons are the most abundant in the human body and how do they differ from the rest of neurons?

II. Types of neurons based on the types of processes

- i) **Unipolar neurons:** They are '**sensory neurons**' having a single process arising from the cell body, which divides into two branches, one of which acts as a **dendrite** and the other as an **axon**. The 'soma' is found in the **dorsal root ganglion** of spinal nerve. They are also called '**pseudounipolar neurons**'.
- ii) **Bipolar neurons** have one axon and one dendrite that directly arise from the cyton. They are found in the **retina** of the eye, **sensory cells** of the internal ear, olfactory sensory epithelium, etc.
- iii) **Multipolar neurons** have one axon and two or more dendrites. Most neurons in our body are **multipolar** neurons.

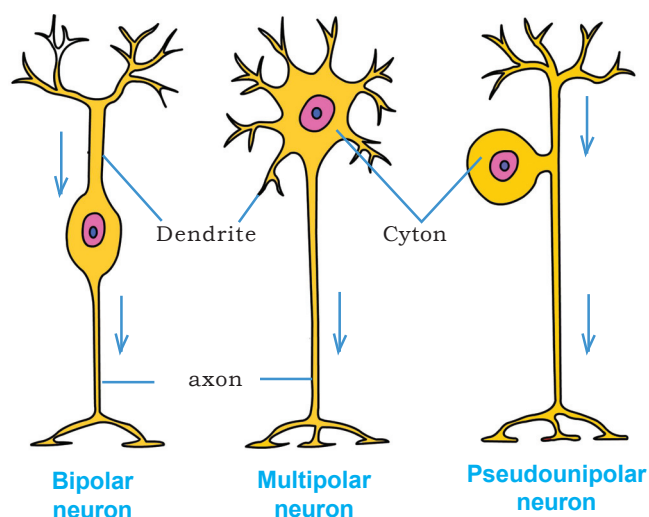


Fig. 2.22 Types of neurons

III. Types of axons based on the presence or absence of myelin sheath

- i) **Myelinated axons:** In a myelinated fibre of the PNS, the plasmalemma of a Schwann cell wraps around an 'internode' of an axon. The layers of the Schwann cell contain '**myelin**' (with a high proportion of lipids). The outermost layer that contains cytoplasm and nucleus is called the **neurilemma**. Some Schwann cells do not contain myelin. In the **CNS**, the glial cells, called 'oligodendrocytes' myelinate axons (*a single oligodendrocyte can myelinate many axons*). The portions of a myelinated nerve fibre (axon) devoid of myelin sheath are called **nodes of Ranvier**. The gap between two adjacent nodes is called an **internode**. Myelinated nerve fibres occur in the white matter of **CNS**, and in most peripheral nerves.
- ii) **Non-myelinated axons** are commonly found in the grey matter of the CNS and post-ganglionic neurons of the autonomic nervous system.

IV. Types of neurons based on their function

- i) **Sensory neurons (afferent neurons)** carry sensory impulses from the receptors to the CNS.
- ii) **Motor neurons (efferent neurons)** carry motor impulses from CNS to the 'effector organs' such as muscles, glands, etc.
- iii) **Interneurons** are present in **CNS** and they connect sensory neurons and motor neurons.

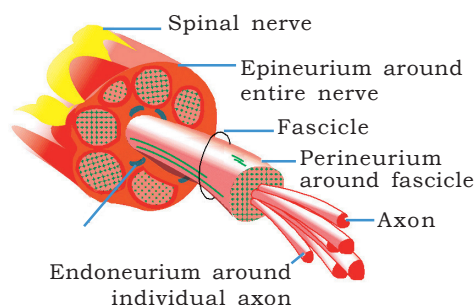


Fig. 2.23 Cross section of a nerve

V. Nerve

A bundle of nerve fibres in the peripheral nervous system form a nerve. Each nerve fibre is covered by a thin connective tissue sheath called **endoneurium**. A group of nerve fibres form a **fascicle** (fasciculus), it is covered by **perineurium**. All the fascicles of a nerve join together to form a bundle, which is surrounded by a connective tissue sheath called **epineurium**, thus forming a '**nerve**'. Nerves are of three types- a) Sensory, b) Motor and c) Mixed.

VI. Neuroglia (supporting cells)

These are the supporting and 'non-conducting' cells that provide a microenvironment suitable for neuronal activity. Unlike neurons, they continue to divide throughout the life. Neuroglial cells of the CNS include

- i. **Oligodendrocytes** (that form myelin sheath as mentioned above);
- ii. **Astrocytes** (star shaped cells) that form interconnected network and bind neurons and capillaries (helping in providing **blood-brain barrier**);
- iii. **Ependymal cells**, which are **ciliated** cells that line the cavities of brain and spinal cord to bring movements in the cerebrospinal fluid;
- iv. **Microglial cells**, which are **phagocytic cells**, of **mesodermal origin**.

Neuroglial cells of the peripheral nervous system include the **satellite cells** and **Schwann cells**. Satellite cells surround the cell bodies in ganglia, and Schwann cells form neurilemma around axons.

Microglia are the only cells of mesodermal origin in the nervous tissue. They form a part of the 'mononuclear phagocytic system' (MNP system). They become activated into 'phagocytes' when there is infection or injury in the nervous system.

VII. Synapse:

The minute structural gap or functional bridge between the axon terminals of one neuron and the dendrites of another neuron is called synapse.

GLOSSARY

Anticoagulants: These are the substances that prevent the coagulation of blood, when blood flows through blood vessels. e.g. heparin

Articular cartilage: It is the cartilage found at the free surfaces of long bones that forms joints

Blubber: It is the thick, fatty layer that lies underneath the skin of whales and other aquatic mammals. It helps in thermal insulation.

Central axis: An imaginary straight line joining the mid point at one end or surface and the midpoint at the opposite end or surface. It is also called the principal axis.

Diapedesis: The passage of leucocytes of blood into the matrix of connective tissues through the walls of capillaries by an amoeboid movement.

Endothelium: The simple squamous epithelium that lines the inner surface of blood vessels and heart.

Epiphyses: These are the expanded ends of long bones, which are made of spongy bone.

Frontal plane: The plane that runs along the anterior – posterior and transverse axes.

Haemocoel: The functional perivisceral cavity, that lies around the visceral organs of arthropods and molluscs which is filled with the blood (haemolymph).

Haversian canal: These are the elongated canals that run parallel to the marrow cavity of compact bones of mammals. They contain blood vessels, lymphatic vessels and nerves.

Hydrostatic skeleton: In pseudo-coelomates and eucoelomates, the body cavity is filled with a fluid which gives proper shape to the body as an endoskeleton does.

Ligament: A dense fibrous tissue strand, which connects bones to other bones in general. Extensive pulling of ligament causes sprain.

Median sagittal plane: The plane that runs along the anterior, posterior and sagittal axes.

Megakaryocytes: Giant cells of red bone marrow that produce blood platelets by fragmentation.

Muscle fatigue: It is the inability of the muscle to undergo further contraction due to the accumulation of lactic acid formed during anaerobic respiration, because of rapid exercise.

Oedema: It is an abnormal accumulation of interstitial fluid beneath the skin or in one or more cavities. Fall in the levels of plasma proteins mainly serum albumin causes a decrease in the osmotic pressure of blood.

Osteoblasts: These are the immature cells that secrete organic matter of the matrix of a growing bone. Immature osteoblasts become mature cells, the osteocytes, in the adult stage.

Perivisceral cavity: It is the cavity that encloses visceral organs. The perivisceral cavity of nematodes is pseudocoelom, that of annelids is eucoelom.

Plane: The flat area that runs through any axis.

Polycythemia: It is an abnormal rise in the RBC count. The people who live at the higher altitudes generally show polycythemia due to low partial pressure of oxygen at the higher altitudes.

Primary induction: It is a kind of interaction between tissues of different origin involved in the development of specialized tissues.

Reticular cells: These are the specialized fibroblasts that secrete reticular fibres of matrix.

Reticular lamina: It is a part of the basement membrane of the epithelium that lies close to the underlying tissues.

Retroperitoneal organs: Organs that are located in between the dorsal body wall and parietal layer of peritoneum. E.g., kidneys

Sagittal axis: An imaginary straight line that joins mid dorsal and mid ventral ends.

Secondary bodycavity: True coelom of eucoelomates is called secondary body cavity, because it is formed by the replacement of primary body cavity, the blastocoel.

Spasms: These are sudden sustained involuntary contractions of smooth muscles of visceral organs.

Stereocilia: These are non-motile cilia-mechanosensing organelles in internal ear, epididymis etc., respond to fluid motion in numerous types of animals for various functions.

Synaptic knobs: These are bulb-like structures formed distally by the axon terminals and contain synaptic vesicles which have neurotransmitters.

Transverse plane: The plane that runs through the sagittal axis and transverse axis.

Vasodilators: These are substances like histamine and bradykinin which increase the capillary permeability and cause inflammation in response to infection or injury.

VERY SHORT ANSWER TYPE QUESTIONS

1. The body of sponges does not possess tissue level of organisation, though it is made up of thousands of cells. Comment on it.
2. What is 'tissue' level of organisation among animals? which metazoans do exhibit this organisation?
3. Animals exhibiting which level of organisation lead relatively more efficient way of life when compared to those of the other levels of organisation? Why?
4. What is monaxial heteropolar symmetry? Name the group of animals in which it is the principal symmetry.
5. Radial symmetry is an advantage to the sessile or slow moving organisms. Justify this statement.
6. What is cephalization? How is it useful to its possessors?
7. Mention the animals that exhibited a 'tube-within-a-tube' organisation

- for the first time? Name their body cavity.
8. Why is the true coelom considered a secondary body cavity?
9. What are retroperitoneal organs?
10. What is enterocoelom? Name the enterocoelomate phyla in the animal kingdom?
11. Distinguish between exocrine and endocrine glands with examples.
12. Distinguish between holocrine and apocrine glands.
13. Mention any two substances secreted by mast cells and their functions.
14. Distinguish between a tendon and a ligament.
15. Distinguish between brown fat and white fat.
16. What is the strongest cartilage? In which regions of the human body, do you find it?
17. Distinguish between osteoblasts, and osteoclasts?

18. Define osteon.
19. What are Volkmann's canals? What is their role?
20. What is a Sesamoid bone? Give an example.
21. What is lymph? How does it differ from plasma?
22. What is the haematocrit value?
23. What are intercalated discs? What is their significance?
24. "Cardiac muscle is highly resistant to fatigue". Justify.
25. Distinguish between '**nucleus**' and '**ganglion**' with respect to the nervous system.

26. Distinguish between '**tracts**' and '**nerves**' with respect to the nervous system.
27. Name the glial cells that form myelin sheath around the axons of central nervous system and peripheral nervous system respectively.
28. Distinguish between white matter and grey matter of 'CNS'.
29. What are microglia and what is their origin and add a note on their function.
30. What are pseudounipolar neurons? Where do you find them?

SHORT ANSWER TYPE QUESTIONS

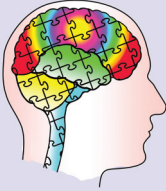
1. Describe the four different levels of organization in metazoans?
2. In which group of bilaterians do you find solid bauplan? Why is it called so?
3. Mention the advantages of coelom over pseudocoelom?
4. Describe the formation of schizocoelom and enterocoelom.
5. Describe briefly about the three types of intercellular junctions of epithelial tissues.
6. Give an account of glandular epithelium.
7. Give a brief account of the cells of areolar tissue.

8. Describe the three types of cartilage.
9. Explain Haversian system.
10. Write short notes on lymph.
11. Describe the structure of a skeletal muscle.
12. Describe the structure of a cardiac muscle.
13. Give an account of the supporting cells of nervous tissue.
14. Describe the structure of a multipolar neuron.
15. Write short notes on (A) Platelets and (B) Synapse.

LONG ANSWER TYPE QUESTIONS

1. What is coelom? Explain the different types of coelom with suitable examples and neat labelled diagrams?
2. What is symmetry? Describe the different types of symmetry in the animal kingdom with suitable examples.
3. Classify and describe the epithelial tissues on the basis of structural modification of cells with examples.

4. Describe the various types of connective tissue proper with suitable examples.
5. What is a skeletal tissue? Describe structure of a compact bone.
6. Give an account of the "formed elements" of Blood?
7. Write notes on striated muscles.



For ignited minds

- Not For Evaluation

1. **Microtome** invented by **His** is used for taking fine sections of tissues for their observation under a microscope.
2. **Canada balsam** is used in the preparation of permanent slides to mount sections of tissues and micro-organisms as its refractive index is same as that of glass.
3. **Collagen** is the most abundant protein in the human body.
4. The **first key transition** in the eumetazoan body plan is the evolution of **tissues**.
5. The **second key transition** in the eumetazoan body plan is the evolution of **Bilateral symmetry**.
6. The **third key transition** in the eumetazoan body plan is the **evolution of coelom**.
7. Structural **complexity** of body organisation is attained due to the evolution of **mesoderm**.
8. The skin and its derivatives constitute the **integumentary system**.
9. The branch of medicine that deals with the study of skin and its diseases is known as **dermatology**.
10. In flatworms the segmentation cavity is completely filled with mesenchyme (**parenchyma** of **mesodermal** origin) whereas in round worms, it is partly filled with it.
11. Dermis of the skin of the large mammals yields leather produced by a chemical treatment called '**tanning**'.
12. Excluding the **haemopoietic tissue**, the most rapidly multiplying tissue is the epithelial tissue. That's why cancers of epithelia (carcinomas) are the most common type of cancers. The more the cells divide, the more is the chance of mutation occurring in the DNA leading to uncontrolled cell cycles.
13. In young people, bones do not break easily and fractures are joined (healed) quickly, where as the reverse is the case in the grownups. Can you make an educated guess on it?
14. When a bone is fully grown, it stops growth as osteoblasts become inactive osteocytes. But, when a bone is fractured how is it healed when there are no osteoblasts?
15. Impulse conduction is relatively faster in the cardiac muscle. Elucidate.
16. When a person slips ascending a stair case, he is likely to develop a sprain. **Can you guess what happened in the tissue concerned.**
17. '**Water, water everywhere but not a drop to drink**' is an old saying. '**Oxygen, oxygen everywhere, but not a pinch to be used for the cell's own respiration**'. What in your opinion are we talking about?
(**CLUE:** It is a type of cell involved in oxygen transport).
18. Passing of certain blood cells through the walls of capillaries into the connective tissue is referred to as **diapedesis**. However, there are some cells which undertake what might be called '**reverse diapedesis**'. Can you guess which cells do that?
(**CLUE:** These cells have a major role to play in the immune system).
19. Can you identify the largest WBC just by looking at the nucleus. If so, what are they and how could you identify?
20. What are the smallest of the 'formed elements' in the human blood? What is their chief role?
(**CLUE:** When you get an intravenous injection, they initiate their role, a few seconds after the syringe is withdrawn).